A Versatile Mixture Distribution and Its Application in Economic Dispatch with Multiple Wind Farms

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***Abstract*—An improved versatile distribution for wind power is proposed, with higher representation accuracy and more suitable applications in economic dispatch (ED) compared with Versatile distribution. Further, versatile mixture distribution (VMD) is proposed to represent all possible wind power distribution with customized arbitrary errors. Compared with Gaussian Mixture distribution (GMD), it has more flexible forms and can represent wind power more accurately. Its cumulative distribution function has analytical forms with higher computational efficiency. Then, real-time dynamic ED model and algorithm based on VMD with multiple wind farms (WFs) is proposed. Based on the VMD model, the probability distributions of wind power of each WF and the summation of all WFs are modeled. The ED model and algorithm solve the problem of the correlation of multiple renewable energy random variables with high computation efficiency for real-time ED. The results show that compared with GMD or unimodal distributions, VMD can represent wind power more accurately and can greatly simplify and speed up the ED model and algorithm. Compared with other methods for multiple renewable energy random variables, the proposed model and algorithm can improve economy significantly.**

***Index Terms*—Dynamic economic dispatch, Gaussian mixture distribution, multiple wind farms, versatile mixture distribution, wind power.**

I. Introduction

RENEWABLE energy resources such as wind power have become a large part of the power system, imposing a great challenge for their uncertainty and many studies have been made on it [1]. However, accurate forecasting of future wind generation across temporal and spatial scales still remains elusive [2]. So it is significant to represent the uncertainty of wind power properly, especially in the area of stochastic economic dispatch (ED) with wind power integration.

To represent the uncertainty of wind power, one effective method is to treat wind power as a conditional probability function, i.e., dividing the forecast wind power into some bins and then modeling the distribution of actual power (or forecast error) within each forecast [3]. Based on this method, many studies have been proposed and can be summarized as follows.

Historical data pairs (forecast power, actual power) are divided by forecast wind power into some forecast bins (FBs), and then the probability density histogram (PDH) in a FB can be regarded as the distribution of actual available wind power when the forecast wind power locates in this FB. The distribution of actual available wind power within each FB can be represented by PDH in [4]–[6]. This method is very precise in theory but it is time-consuming since the PDH is essentially equivalent to the probability distribution of discrete random variables. Discrete calculation results in an increase of the number of variables, thus reducing the computation speed [7]. The scheduled plan should be determined in few minutes or even seconds in real-time economic dispatch (RTED) with wind power integration and the PDH method cannot meet the requirement of calculation efficiency. In addition, a large number of variables brought by PDH will greatly increase the complexity of data processing of wind power. So it can only be applied in the situation where wind power is not easy to represent by a certain distribution model and there are no strict requirements on calculation efficiency.

Due to the above defects of PDH, more commonly used methods are assuming wind power follows certain distribution models. For instance, Gaussian distribution (GD) [8]–[13] has been widely used in earlier studies, so has Beta distribution (BD) [3], [14]–[16], and Versatile distribution (VD) [7] shows better representation effect than other models. However, with the gradual expanding scale of the wind farm (WF), the probability distribution of wind power is more and more complicated due to meteorological and geographical correlations and one important feature is the characteristics of "multi peaks." In addition, multiple WFs need to be considered in the integration of wind power at high-penetration and different access points. The probability distribution of the summation of all WFs (i.e., sum WF) outputs also has the characteristic of "multi peaks" due to the meteorological and geographical correlations of each WF in it. However, all the above distribution models are "unimodal" (neglecting the curve form of "U" of BD). Ma [6] directly stated that the WF (i.e., the WF being studied in this paper) does not follow the above distribution models.

In summary, due to the characteristics of "multi peaks" in the wind power probability distribution, large representation errors would be generated if the above distribution models (i.e., GD, BD and VD) are employed. Thus the scheduled plan would not

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be determined accurately in ED, having huge impact on the economic operation and security of the power system with wind power integration. How to well represent wind power is basic in ED with multiple WFs and a new distribution model that can represent any form of wind power and is also suitable for ED with wind power integration is needed.

It is worth to mention that as a classical model of data analysis, Gaussian Mixture distribution (GMD) is widely used in the areas of machine learning [17], pattern recognition [18] and image processing [19]. It is also widely used in the field of the power system [20]–[23], and has begun to be used in the representation of wind power these years [24]. Weighted combination of multiple GDs in GMD can be used to fit a variety of probability distributions, so it shows a high potential to represent wind power with any distribution form. Due to similar requirement, Weibull mixture model is also presented on energy systems [25]. However, Weibull model is mainly employed to represent the wind speed and is not as widely used as GMD. GMD is regarded as a comparison object in this paper. The new distribution model proposed in this paper shows better representation effect for wind power and more application advantages in ED.

In order to integrate wind power into power system, the traditional economic dispatch (ED) method is increased pressure to improve. Due to the variability and uncertainty, high penetration of wind power is expected to result in significant operational challenges [26]. Especially, the integration of wind power at high-penetration levels into power system makes it impossible to attain an acceptable load and generation balance. To this end, new economic dispatch methods with high penetration of wind power are required. Currently, many methods have been proposed to solve the ED with wind power participation. Simple method is deterministic approaches [27]–[28], which treat wind generation as negative load. In the conventional ED, the regulation reserves only need to mitigate the effects of load forecast errors. However, wind power is far more variable and unpredictable than loads [29]. The forecast error of wind power could be 15%–20% in typical day-ahead forecasts [30] while the forecast error of loads is only 1%–3% in typical load forecasts [31]. Hence, as shown in [32]–[34], there is an urgent need to revisit the approach to set regulation reserves [35]. So deterministic ED approaches are not suitable for wind power participation at high-penetration levels into power system.

Robust approaches is another method in ED, in which wind power falls within a given uncertainty set [34], [36]–[42]. Within the given uncertainty set, robust approaches search for a solution that can ensure system robustness. Therefore, the robust approaches are often considered to be conservative [35].

Stochastic ED uses a probabilistic distribution model to represent the uncertainty of wind power, offering better mechanisms to manage the uncertainty explicitly. One of the key advantages is that distribution model enables system operators to maintain an acceptable level of risk [35]. As shown in [43] and [44], stochastic ED based on distributional forecasts can improve the system efficiency in terms of reducing system reserves. Scenario-based stochastic programming has been utilized to address the uncertainty and variability of wind power in stochastic ED in [10], [32], [44]–[46]. Markov chain models [47]–[50] have been proposed to model the wind power and then are used in stochastic ED [35]. A Versatile distribution model is formulated in [7], simplifying the wind uncertainties for its cumulative distribution function (CDF) and inverse CDF have analytical forms, making it possible to deal with stochastic ED by classical method.

However, almost all the above papers cannot deal with multiple wind farms (WFs) accurately, because the correlation of different WFs makes it different to set system reserves accurately. In addition, since classical algorithms have been proven efficient and reliable with a lot of successful experience, there still exists a pressing need to develop the ED algorithm with wind power integration in the classical fashion [7]. This is what this paper focuses on.

The major contributions of this paper are summarized as follows:

1) Based on the research results in [7], an Improved Versatile distribution (IVD) is proposed in this paper, with more accurate representation effect compared with VD. Then based on IVD, Versatile Mixture distribution (VMD) is proposed to represent any distribution form of wind power. Compared with GMD, it has more flexible forms and its cumulative distribution function (CDF) has analytical forms. Thus, it has higher computational efficiency and it can represent wind power much more accurately.
2) Based on VMD, a dynamic real-time economic dispatch (RTED) model and algorithm with multiple WFs is proposed. Compared with other ED models with wind power integration, the proposed model based on VMD can represents all possible wind power distribution with high efficiency. The proposed model takes the effect of correlation of multiple WFs variables into account for the system regulation reserves reasonably, meeting the need of economy in power system. An economic operation interval (EOI) fitting method is proposed to increase wind power representation accuracy and decrease the number of mixture components of VMD. The accuracy and the computation efficiency of the ED model and algorithm can be improved by the EOI fitting method.

The remaining of this paper is organized as follows. In Section II, wind power conditional distribution model based on IVD is proposed and its representation advantage compared with VD is analyzed. In Section III, VMD is proposed to represent any distribution form of wind power. In Section IV, the RTED model with multiple WFs that considers the correlation of multiple wind power variables based on VMD is proposed. In Section V, an efficient algorithm named equal incremental fuel cost - quadratic programming - sequential linear programming (EIFC-QP-SLP) is proposed to meet the real-time need. An EOI fitting method is proposed to improve the accuracy and the computational efficiency of the ED model and algorithm. In Section VI, the representation effect and computational efficiency of VMD are compared with those of GMD using actual historical wind power data. Numerical experiments of RTED are conducted using IEEE 30-bus system with two WFs. Section VII provides conclusions.

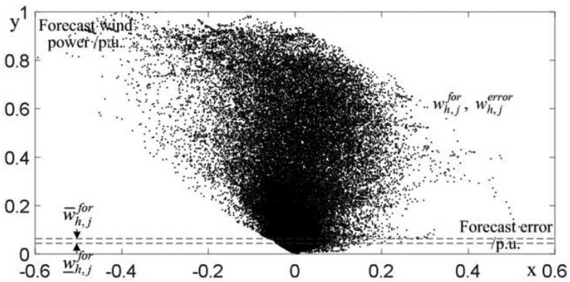

Fig. 1. Typical historical data of forecast wind power and corresponding forecast error.

II. Wind Power Conditional Distribution Model Based on IVD

A. Historical Data and Forecast Bins

The historical data set is employed to get the conditional distribution of wind power. All the data pairs $w_{h,j}^{for}, w_{h,j}^{act}$ are normalized, hence varying within [0, 1]. $w_{h,j}^{for}$ is historical forecast wind power of *j*-th WF and $w_{h,j}^{act}$ is corresponding historical actual wind power. The corresponding forecast error $w_{h,j}^{err}$ is defined as $w_{h,j}^{act} - w_{h,j}^{for}$. The set of historical data pair $w_{h,j}^{for}, w_{h,j}^{error}$ is shown in Fig. 1 and each data pairs $w_{h,j}^{for}, w_{h,j}^{error}$ is a point in it.

The forecasting domain (i.e., [0, 1] p.u.) is divided into certain number of bins with certain number of data pairs $w_{h,j}^{for}, w_{h,j}^{error}$ in it. Each bin is called a "forecast bin" (FB) in this paper. In this paper, FB is designed as the following steps.

Step 1: Sorting all the data pairs $w_{h,j}^{for}, w_{h,j}^{error}$ from small to large by the forecast wind power $w_{h,j}^{for}$. Assuming the number of historical data pairs is *S*.

Step 2: The lower bound of FB 1 is 0 p.u. Determining the upper bound of FB 1 to cover at least *K* data pairs in Fig. 1. *K* is a user-defined integer according to the number of historical data pairs *S*.

Step 3: From $m1 = 1, 2, \ldots, M1$, setting the upper bound of FB $m1$ as the lower bound of FB $m1 + 1$. Determining the upper bound of FB $m1 + 1$ to cover at least *K* data pairs in Fig. 1 until all the historical data are used. Then we can get $M1$ FBs.

The program of designing FBs is to take out the data pairs in the FB from historical data set (Fig. 1). The range of FB is recorded as $[\underline{w}_{h,j}^{for}, \bar{w}_{h,j}^{for}]$ and the set of data pairs in this FB is shown in Fig. 2(a). The set of forecast error data in this FB can be divided by a certain quantity of bins ($M2$) to get the PDH, as shown in Fig. 2(b).

B. Wind Power Conditional Distribution Model Based on IVD

Wind power conditional distribution model is based on IVD. The PDF, CDF and inverse CDF of IVD are as follows, respectively.

$$f(w_{j,t}) = \frac{\alpha\beta e^{-\alpha(w_{j,t}-\gamma-w_{j,t}^{for})}}{\left(1+e^{-\alpha(w_{j,t}-\gamma-w_{j,t}^{for})}\right)^{\beta+1}} \quad (1)$$

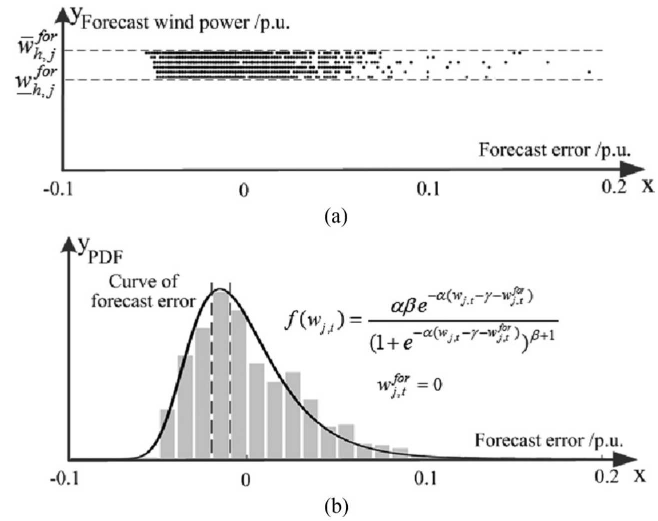

Fig. 2. IVD of wind power forecast error in the FB. (a) Set of data pairs in the FB. (b) PDH and IVD of forecast error in the FB.

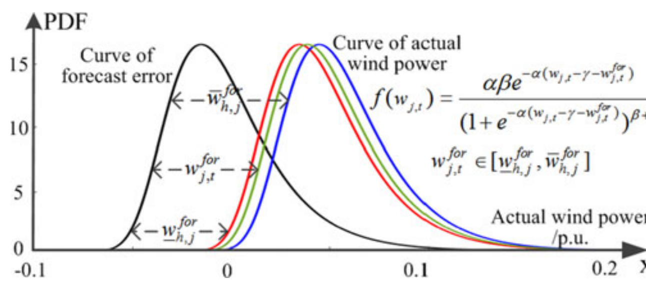

Fig. 3. Actual wind power modeled by IVD in the FB.

$$F(w_{j,t}) = (1 + e^{-\alpha(w_{j,t}-\gamma-w_{j,t}^{for})})^{-\beta} \quad (2)$$

$$F^{-1}(c) = \gamma - w_{j,t}^{for} - \frac{1}{\alpha}\ln(c^{-1/\beta} - 1) \quad (3)$$

where α, β, and γ are the shape parameters of IVD, $\alpha > 0$, $\beta > 0$, $-\infty < \gamma < +\infty$, $w_{j,t}^{for}$ is the forecast power of the *j*-th WF at time *t*, which can be obtained in ED. That is to say that the actual power modeled by IVD can be got by the IVD curve of forecast error of wind power moving to right on the abscissa by $w_{j,t}^{for}$. Then before modeling the actual wind power, we need to model the distribution of forecast error according to different forecast wind power. We use the IVD ($w_{j,t}^{for} = 0$) to model the curve of forecast error of wind power in each FB by PDH fitting, as shown in Fig. 2.

For each FB $[\underline{w}_{h,j}^{for}, \bar{w}_{h,j}^{for}]$, we can get the distribution of wind power forecast error by IVD, i.e., the group of parameters in (1)–(3). In ED, forecast wind power of each WF at each time interval $w_{j,t}^{for}$ can be obtained. Combined by the group of parameters in the FB where the forecast wind power located in, i.e., $w_{j,t}^{for} \in [\underline{w}_{h,j}^{for}, \bar{w}_{h,j}^{for}]$, the distribution of actual wind power by IVD in (4)–(6) can be obtained, as shown in Fig. 3. The black line is the distribution of forecast wind power and it is same with the black line in Fig. 2. The green line is the distribution of actual wind power by IVD, i.e., black line move along the x-axis by

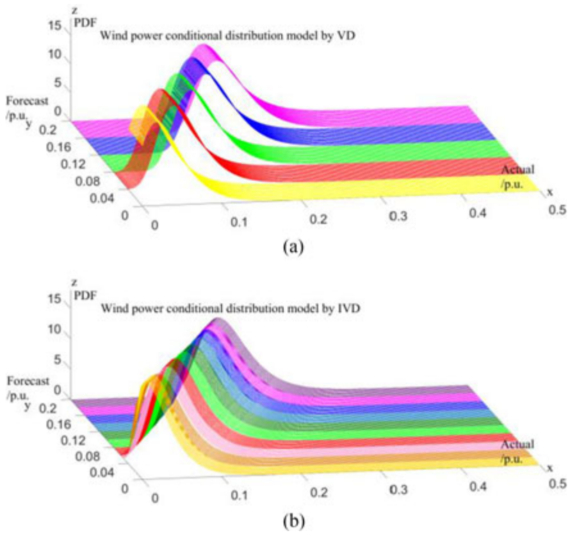

Fig. 4. Wind power conditional distribution models by VD and IVD (forecast wind power: 0–0.2 p.u.). (a) Model by VD. (b) Model by IVD.

$w_{j,t}^{for}$. The distribution of actual wind power in this FB is located between the red line and blue line according to different $w_{j,t}^{for}$.

The distribution of actual wind power based on IVD is essentially a conditional model. Groups of shape parameters α, β, and γ in IVD model (1)–(3) in each FB are obtained by historical data set as the process in Figs. 1 and 2. FBs and the group of shape parameters α, β, and γ in this FB are recorded in the offline look-up table. Then in ED, wind power conditional distribution model can be obtained by $w_{j,t}^{for}$. The shape parameters can be obtained by the FB where $w_{j,t}^{for}$ located in i.e., $w_{j,t}^{for} \in [\underline{w}_{h,j}^{for}, \bar{w}_{h,j}^{for}]$. The conditional distribution of actual wind power modeled by IVD can be got by the IVD curve of forecast error in this FB moving to right on the abscissa by $w_{j,t}^{for}$, as shown in Fig. 3.

C. Comparison Between IVD and VD

IVD and VD models are all employed to model the distribution of actual wind power. The most essential difference between IVD and VD is the PDHs that they fit, i.e., forecast error and actual power, respectively. Fig. 4 shows wind power conditional distribution models by VD and IVD when forecast wind power located in $0 \sim 0.2$ p.u.

1) For IVD model, the PDF of forecast error is same in the same FB while the PDF of actual power is the probability distribution of forecast error moving to right on the abscissa by $w_{j,t}^{for}$. The forecast power is introduced as a parameter in the wind power model. Compared with VD that employ one distribution to model the actual wind power in each FB (as shown in Fig. 4(a)), IVD that employs a series of distribution (as shown in Fig. 3 and Fig. 4(b)) greatly reduce the error of system reserves capacity and ramps capacity in ED based on VD.
2) The IVD model has a sound bin designing method. The FBs of IVD are designed unevenly. For the reason that the distribution of historical data set is uneven as shown in Fig. 1, the number of FBs *M1* of IVD can be much larger compared with VD, as shown in Fig. 4, greatly increasing the accuracy of representation effect of wind power.
3) The probability distribution model of actual power got by curve moving of IVD has the same expression as VD, so IVD has the same mathematics advantage of analytical forms in ED.

III. VMD Model

As mentioned above, with the gradual expanding scale of the WF, the probability distribution of wind power shows more complex forms due to meteorological and geographical correlations, and one important feature is a characteristic of "multi peaks". At this time, GD, BD and IVD are unable to fit accurately. Based on IVD, a new distribution model named VMD is proposed to represent any distribution form of wind power and have good mathematical properties.

A. Introduction of VMD Model

The PDF and CDF of VMD for wind power are as follows, respectively.

$$f(w_{j,t}) = \sum_{i=1}^{L} k_i \frac{\alpha_i \beta_i e^{-\alpha_i (w_{j,t} - \gamma_i - w_{j,t}^{for})}}{\left(1 + e^{-\alpha_i (w_{j,t} - \gamma_i - w_{j,t}^{for})}\right)^{\beta_i + 1}} \tag{4}$$

$$F(w_{j,t}) = \sum_{i=1}^{L} k_i \left(1 + e^{-\alpha_i (w_{j,t} - \gamma_i - w_{j,t}^{for})}\right)^{-\beta_i} \tag{5}$$

where L is the number of mixture components and k_i is the weight of the i-th mixture component, subject to $0 < k_i \leq 1$ and $\sum_i^L k_i = 1$. α_i, β_i and γ_i are the shape parameters of the i-th mixture component of IVD, $\alpha > 0$, $\beta > 0$, $-\infty < \gamma < +\infty$.

With a suitable number of mixture components and shape parameters of each component, wind power with any distribution form can be represented within a certain error tolerance. VMD is a more general distribution model of wind power and IVD can be deemed as a special case where the number of mixture components is 1. Just like IVD model, before modeling the actual wind power, we need to model the distribution of forecast error, i.e., using the VMD ($w_{j,t}^{for} = 0$) to model the curve of forecast error of wind power in each FB by PDH fitting.

The actual wind power or forecast error often follows irregular distribution due to the meteorological and geographical correlations in WFs. Take the whole regions of Ireland's power system with 53,660 historical data pairs (2014.9.5-2016.3.16) [51] for instance, we can get the forecast error PDH by the bin designing method of IVD and the forecast error PDH in part FBs are as shown in Fig. 5. It can be shown that "multi peaks", "fat tails" are detected in the forecast error PDH in some FBs, which cannot be represented accurately by GD, BD and IVD.

It is worth noting that for the PDF curve in a FB, the number of bins of forecast error *M2* is also very important. The curve shape and the number of peaks may have larger difference for different *M2*. From the aspect of PDH accuracy, larger *M2* results in accurate representation effect, but if *M2* is too large, the PDH

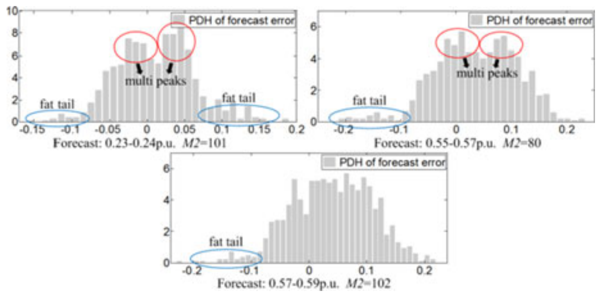

Fig. 5. Wind power forecast error PDH in part FBs of the whole regions of Ireland's power system.

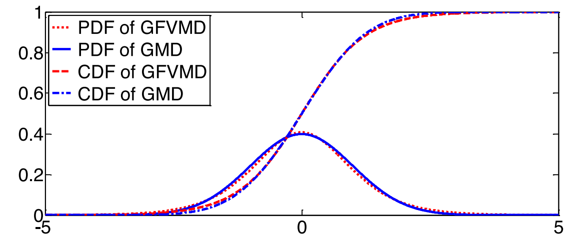

Fig. 6. Comparison of PDF and CDF of GFVMD and GMD.

will be difficult to fit, so the *M2* should be large enough but meet the condition for curve fitting. A reasonable method is first selecting a larger value and then gradually decreasing the value, until the curve not sparse under the premise of the same curve trend. Detailed examples can be seen in the case study below.

However, for the CDF curve, the cumulative distribution histogram (CDH) of wind power forecast error will not be sparse with the increase of *M2*. Since the CDH is more accurate when *M2* is larger, *M2* of the CDF curve can be as large as possible. This difference will result in more accurate CDH of the CDF fitting method compared with the PDF fitting method for wind power.

B. The Representation Effect Analysis of VMD

To fit this irregular distribution, a classic distribution model is Gaussian Mixture distribution (GMD), which is a linear combination of finite GDs [21]. Similar to taking GD as a comparison object for IVD, GMD can be regarded as the comparison object for VMD.

The PDF of GMD for wind power is as follows.

$$f(w_{j,t}) = \sum_{i=1}^{L} k_i \frac{1}{\sqrt{2\pi}\sigma_i} e^{-\frac{(w_{j,t}-\mu_i-w_{j,t}^{for})^2}{2\sigma_i{}^2}} \tag{6}$$

where L is the number of mixture components and k_i is the weight of the *i*-th mixture component, subject to $0 < k_i \leq 1$ and $\sum_i^L k_i = 1$. u_{i} and σ_{i} are the mean and variance of the *i*-th mixture component of GVD, $-\infty < u_i < +\infty$, $\sigma_i > 0$.

1) The Representation Effect of VMD and GMD: Since the PDF of IVD has the property of distortion, the curve is much more flexible than GD whose PDF is symmetrical [7]. IVD and GD are the special case where $L = 1$ of VMD and GMD, respectively. So compared with GMD, VMD shows better representation effect with the same *L*.

On the other hand, to obtain the parameters of GMD, traditional approaches, such as least square algorithm and expectation maximization algorithm, use the PDF of GMD to fit the PDH. However, it should be noticed that it is the CDF, not the PDF that determines the statistical model of wind power in ED [7]. Although the accuracy of PDF fitting is a sufficient condition for the accuracy of CDF fitting if the number of components is big enough, errors cannot be ignored when the number of components decreases. In some worse cases, the representation error of CDF can be accumulated in the process of PDF integration, reducing the representation accuracy. Since the CDF of IVD and VMD has analytical forms, the parameters of the CDF curve can be obtained easily by least squares fitting (e.g., toolbox of "cftool" in matlab). On the contrary, for distribution models without analytical forms of CDF such as GD, BD and GMD, only the parameters of PDF can be obtained and PDF integration has been employed to obtain the CDF curve. Then the representation error of CDF of GMD will increase. On the other hand, the CDH fitting method is more accurate than the PDF fitting method for wind power since the CDH is more accurate when *M2* is larger.

The representation effect of CDH determines the representation effect of distribution model for wind power. Therefore for the above three reasons, VMD that has more accurate representation effect shows better representation effect compared with GMD with the same *L*.

2) Computation Efficiency of VMD and GMD: IVD has much better computation efficiency for CDF than GD [7]. Therefore, VMD has much better computation efficiency for CDF than GMD with the same *L*. Therefore, VMD has much better computation efficiency for CDF than GMD within a certain error tolerance.

For inverse CDF, both VMD and GMD have no analytical forms and iterative calculation is required. Calculation of CDF is required in each iteration so VMD also has much better computation efficiency for inverse CDF than GMD.

C. Gaussian Form of VMD

If we substitute the parameters α_i, β_i, γ_i and k_i with $1.63/\sigma_i$, 1, μ_i and k_i into (4) and (5), then the Gaussian form of VMD (GFVMD) can be obtained. The PDF and CDF of GFVMD are as follows.

$$f(w_{j,t}) = \sum_{i=1}^{L} k_i \frac{\frac{1.63}{\sigma_i} e^{-\frac{1.63(w_{j,t}-\mu_i-w_{j,t}^{for})}{\sigma_i}}}{\left(1+e^{-\frac{1.63(w_{j,t}-\mu_i-w_{j,t}^{for})}{\sigma_i}}\right)^2} \tag{7}$$

$$F(w_{j,t}) = \sum_{i=1}^{L} k_i \left(1+e^{-\frac{1.63(w_{j,t}-\mu_i-w_{j,t}^{for})}{\sigma_i}}\right)^{-1}. \tag{8}$$

The PDF and CDF curve of GFVMD and GMD ($L = 1$) are as shown in Fig. 6. The GMD in Fig. 6 is standard GD with the parameter $\mu = 0$, $\sigma = 1$. The maximum absolute difference of PDF (probability density) of GFVMD compared with GMD is 0.0187 and the maximum absolute difference of CDF (probability) of GFVMD compared with GMD is 1.46%. It is worth mentioning that the maximum absolute difference of

CDF remains to be 1.46% when the GMD ($L = 1$) in Fig. 6 is not standard GD.

If the number of mixture components $L > 1$, the difference of PDF and CDF of GFVMD compared with GMD will be reduced and smooth. The PDF and CDF of GDVMD and GMD will be closer. In practice, GDVMD can approximate or substitute GMD in high accuracy.

Since the studies of GMD have been more mature [20]–[24], within a certain error tolerance, GFVMD can use the studies of GMD. GFVMD can be achieved with almost equal results in characterizing uncertainty compared with GMD, but GFVMD has better mathematical properties that CDF has analytical forms, and has much faster computation efficiency than GMD, which gives GFVMD more application potential than GMD.

IV. RTED With Multiple WFs Model Based on VMD

A. RTED Model With Multiple WFs That Considers the Correlation of Multiple Wind Power Variables

The scheduled powers of interval $t = 1 \ldots T$ are obtained by the newest load and wind power forecast values of T intervals (e.g. 5 min) in RTED, minimizing the total cost under the constraints. RTED model with multiple WFs that considers the correlation of multiple wind power variables is as follows [52], [53].

$$\begin{aligned}\min : f &= \sum_{t=1}^{T} f_t = \sum_{t=1}^{T} (f_{g,t}+f_{w,t})\\ &= \sum_{t=1}^{T}\left[\sum_{i=1}^{I} C_{g,i,t}(p_{i,t}) + \sum_{j=1}^{J} C_{w,j,t}(w_{j,t})\right]\end{aligned} \tag{9}$$

where $f_t,\ f_{g,t},\ f_{w,t}$ are the total generation cost, thermal generators (TGs) cost, WFs cost at time *t*, respectively; to facilitate the discussion, each WF is referred to as WF-I, WF-II ... WF-*j*, the sum power of all WFs (i.e., sum WF) is referred to as WF-SUM; $p_{i,t}$ is the scheduled power of the *i*-th TG at time *t*; $w_{j,t}$ is the scheduled wind power of WF-*j* at time *t*; *I* and *J* is the total number of TGs and WFs, respectively.

The TGs cost at time *t* in (9) can be calculated by

$$f_{g,t} = \sum_{i=1}^{I}[a_i p_{i,t}^2 + b_i p_{i,t} + c_i] \tag{10}$$

where a_i, b_i and c_i are the quadratic cost coefficient of the *i*-th TG, respectively.

The WFs cost at time *t* in (9) can be calculated by [52], [53]

$$\begin{aligned}f_{w,t} = \sum_{j=1}^{J} &[d_j w_{j,t} \\ &+ k_{un,j} \int_{w_{j,t}}^{w_{r,j}} (w_{av,jt} - w_{j,t}) f_j(w_{av,j,t}) dw_{av,j,t}\\ &+ k_{ov,j} \int_{0}^{w_{j,t}} (w_{j,t} - w_{av,j,t}) f_j(w_{av,j,t}) dw_{av,j,t}]\end{aligned} \tag{11}$$

where d_j is the operating cost of WF-*j*; $k_{un,j}$ is the underestimation cost of WF-*j*; $k_{ov,j}$ is the overestimation cost of WF-*j*; $w_{av,j,t}$ is the actual available power of WF-*j* at time *t*.

Subject to [52], [53]:

$$\sum_{i=1}^{I} p_{i,t} + \sum_{j=1}^{J} w_{j,t} = L_t, \forall t \tag{12}$$

$$p_{\min,i} \le p_{i,t} \le p_{\max,i}, \forall i, t \tag{13}$$

$$0 \le w_{j,t} \le w_{r,j}, \forall j, t \tag{14}$$

$$\begin{aligned}&p_{i,t} - p_{i,t-1} \le r_{u,\max,i}, \forall i, t\\ &p_{i,t-1} - p_{i,t} \le r_{d,\max,i}, \forall i, t\end{aligned} \tag{15}$$

$$\begin{aligned}&0 \le r_{u,i,t} \le \min\{p_{\max,i} - p_{i,t}, r_{u,\max,i}\}, \forall i, t\\ &0 \le r_{d,i,t} \le \min\{p_{i,t} - p_{\min,i}, r_{d,\max,i}\}, \forall i, t\end{aligned} \tag{16}$$

$$\begin{aligned}&\Pr\left\{\sum_{i=1}^{I} r_{u,i,t} \ge \sum_{j=1}^{J}(w_{j,t} - w_{av,j,t})\right\} \ge c_u, \forall t\\ &\Pr\left\{\sum_{i=1}^{I} r_{d,i,t} \ge \sum_{j=1}^{J}(w_{av,j,t} - w_{j,t})\right\} \ge c_d, \forall t\end{aligned} \tag{17}$$

$$-\mathbf{F}^{\max} \le \mathbf{F}_t \le \mathbf{F}^{\max}, \forall t \tag{18}$$

where

(12) is the supply-demand balance constraint; L_t is the forecasted power demand at time *t*;

(13) is the TGs capacity constraint; $p_{\max,i}$ and $p_{\min,i}$ are the upper and lower generation limit of the *i*-th TG, respectively;

(14) is the WFs capacity constraint; $w_{r,j}$ is the capacity limit of WF-*j*;

(15) is the TGs ramp-rate constraint; $r_{u,i,t}$ and $r_{d,i,t}$ are the available amount of up and down regulation reserves provided by the *i*-th TG, and $r_{u,\max,i}$ and $r_{d,\max,i}$ are the maximum amount of up and down regulation reserves that the *i*-th TG is capable of providing within a specific time period (e.g., 5 min), respectively;

(16) and (17) are the regulation reserves constraint; c_u and c_d are the confidence levels for having sufficient up and down regulation reserves, respectively; the load forecast error is ignored;

(18) is the transmission constraint; $\mathbf{F}_t$ is the transmission power on a power line at time *t*; $\mathbf{F}^{\max}$ is the transmission power limit on a power line; direct current power flow is employed for the transmission constraint in this paper.

The classical algorithm in ED such as EIFC method and SLP method has been proven reliable and effective with a lot of successful experience. To employ the SLP method, the partial derivatives of objective function with respect to the scheduled wind power in (10) and (11) need to be computed in the vicinity of an operating point. The derivatives of objective function are as follow.

$$\frac{\partial f_{g,t}}{\partial p_{i,t}} = \sum_{i=1}^{I}[2a_i p_{i,t} + b_i] \tag{19}$$

$$\frac{\partial f_{w,t}}{\partial w_{j,t}} = \sum_{j=1}^{J}[d_j - k_{un,j} + (k_{un,j} + k_{ov,j})F_{j,t}(w_{j,t})] \tag{20}$$

where $F_{j,t}(w_{j,t})$ is the CDF of the WF-j output for the forecast $w_{j,t}$ that locates in the forecast bin $m1_j$.

The chance constraints in (17) can be converted to linear and deterministic ones in the SLP-based algorithms as follows [52], [53]

$$\sum_{j=1}^{J} w_{j,t} - \sum_{i=1}^{I} r_{u,i,t} \leq F_{\Sigma,t}^{-1}(1-c_u), \forall t$$
$$-\sum_{j=1}^{J} w_{j,t} - \sum_{i=1}^{I} r_{d,i,t} \leq -F_{\Sigma,t}^{-1}(c_d), \forall t \quad (21)$$

where $F_{\Sigma,t}^{-1}(1-c_u)$ and $F_{\Sigma,t}^{-1}(c_d)$ are the inverse CDF of the sum WF output for the WF-SUM forecast power $w_{sum,t}$ that locates in the forecast bin $m1_{SUM}$.

B. Processing of the Correlation of Multiple Wind Power Random Variables

The wind power uncertainties in power system stochastic ED are embodied in the CDF of each WF (i.e., WF-j) within the corresponding forecast bin in (20) and the inverse CDF of the sum WF (i.e., WF-SUM) within the forecast bin of the sum WF in (21). If there is only one WF in the power system, the wind power distributions in (20) and (21) are same. While if there are multiple WFs in the power system, the wind power distributions of each WF and sum WF must be modeled separately. The correlation of multiple wind power random variables are embodied in the wind power distributions of each WF and sum WF. This is the solving difficulties for this model. Even if the wind power distribution of each WF can be represented by a certain distribution model such as GD, BD or IVD, the wind power distribution of sum WF cannot be modeled by the same distribution model due to the meteorological and geographical correlations of each WF in the power system. That is to say it is difficult to find a simple distribution model to represent the wind power distribution WF-SUM. In addition, for a WF with large-scale, it can also be regarded as the sum WF that consists of several internal small WFs, the wind power distribution can also show arbitrary forms. Take GD for instance, even if each WF is assumed to follow GD, the wind power distribution of sum WF does not necessarily follow GD. Actually, the wind power distribution of sum WF can also show arbitrary forms with different correlations.

To solve the above problem, one approach is that all WFs are regarded as one WF (i.e., sum WF) no matter how many WFs there are in the power system [42]. However, there are two defects for this approach. The first one is that it is difficult to get the scheduled power of each WF after the scheduled power of sum WF is obtained. The second one is that the transmission constraints cannot be satisfied for the transmission constraints are related to the accessing points of each WF in the power system. Another approach is that only the uncertainties of each WF are considered and the regulation reserves are set respectively [54]–[55]. However, due to the reason that the correlation of each WF is not considered, the regulation reserves could be excessive. Therefore, the stochastic dynamic ED with wind

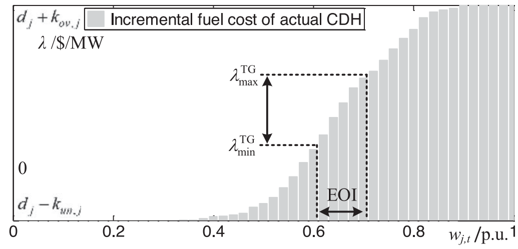

Fig. 7. Incremental fuel cost of actual CDH of WF.

power integration model must be solved by the accurate representation of the wind power distributions of each WF (WF-I, WF-II ... WF-j ... WF-J) and sum WF (WF-SUM).

Based on the VMD model, the wind power cumulative distribution histograms (CDHs) of each WF and the sum WF can be fitted accurately. Therefore, the problem of the correlation of multiple WFs in (21) can be solved by the modeling of the sum WF by VMD. The wind power distribution of each WF in (20) can also be solved by modeling with VMD. With this method, both the wind power distributions of each WF and their correlations can be considered accurately in stochastic ED with multiple WFs.

As analyzed in Section III, the CDF of VMD has analytical form. Compared with GMD, it can represent wind power more accurately and its CDF and inverse CDF has higher computational efficiency. Based on VMD, the stochastic dynamic RTED can be solved by reliable classical algorithms such as EIFC method and SLP method and the computational efficiency can meet the real-time need. Detailed comparison and analysis can be seen in the case study in this paper.

C. Analysis of Wind Power Cost and Wind Power Economic Operation Interval (EOI) Fitting Method

The incremental fuel cost (λ) of WF-j can be written as (22) by (20). Compared with TGs, the operating cost of WF is very low while the underestimation and overestimation cost are relatively very high [40], [42]. So the range of incremental fuel cost of TGs is much narrower than that of WFs. The range of incremental fuel cost of TGs is defined as $[\lambda_{\min}^{\mathrm{TG}}, \lambda_{\max}^{\mathrm{TG}}]$.

$$\begin{aligned}\frac{\partial C_{w,j,t}(w_{j,t})}{\partial w_{j,t}} &= d_j - k_{un,j} F_{j,t}(w_{\max,j}) - k_{ov,j} F_{j,t}(0) \\ &\quad + (k_{un,j} + k_{ov,j}) F_{j,t}(w_{j,t}) \\ &= d_j - k_{un,j} + (k_{un,j} + k_{ov,j}) F_{j,t}(w_{j,t}). \quad (22)\end{aligned}$$

The curve of (22) is as shown in Fig. 7. λ_j stands for the incremental fuel cost of actual CDH of WF-j, which is within $[d_j - k_{un,j},\ d_j + k_{ov,j}]$. The range of incremental fuel cost of TGs is $[\lambda_{\min}^{\mathrm{TG}}, \lambda_{\max}^{\mathrm{TG}}]$ in Fig. 7. We can model the above CDH accurately by VMD with three mixture components ($L = 3$) as shown in Fig. 8.

If the incremental fuel cost of WFs locates outside $[\lambda_{\min}^{\mathrm{TG}}, \lambda_{\max}^{\mathrm{TG}}]$, the incremental fuel cost of the power system will also locates outside $[\lambda_{\min}^{\mathrm{TG}}, \lambda_{\max}^{\mathrm{TG}}]$, then the scheduled power of TGs will be all on the upper or lower generation limit. At this

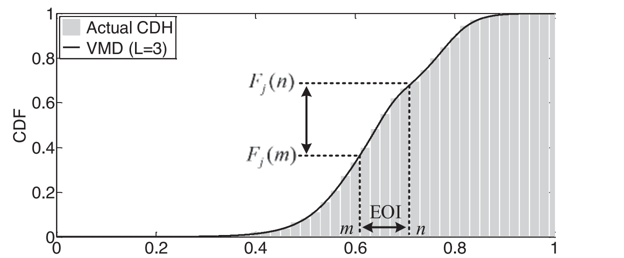

Fig. 8. Actual CDH of WF and CDF of VMD with three mixture components.

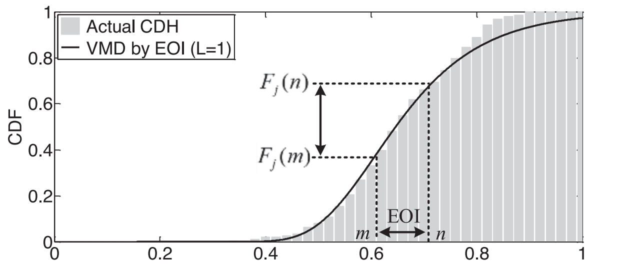

Fig. 9. Actual CDH of WF and CDF of VMD by EOI fitting method.

time, the scheduled power of WFs is determined by the system constraint rather than economic principles. The accurate representation for wind power has no influence for the scheduled result of the system. So the representation effect for wind power CDH whose incremental fuel cost locates outside $[\lambda_{\min}^{\mathrm{TG}}, \lambda_{\max}^{\mathrm{TG}}]$ has no meaning as shown in Fig. 7. From the economic view, it is not necessary to fit the whole range of CDH of WFs. To this end, the EOI is defined as [*m*, *n*] to stand for the necessary interval that wind power CDH needs to be represented accurately as shown in Fig. 8. The interval of incremental fuel cost of WFs in EOI should correspond to the interval of incremental fuel cost of all TGs in this system $[\lambda_{\min}^{\mathrm{TG}}, \lambda_{\max}^{\mathrm{TG}}]$. Combined with (22), The CDF value of the boundary of EOI as shown in Fig. 8 can be got by

$$\begin{aligned} F_j(m) &= (\lambda_{\min}^{\mathrm{TG}} - d_j + k_{un,j})/(k_{un,j} + k_{ov,j}) \\ F_j(n) &= (\lambda_{\max}^{\mathrm{TG}} - d_j + k_{un,j})/(k_{un,j} + k_{ov,j}). \end{aligned} \tag{23}$$

Therefore, just by the accurate representation in EOI of WFs in Fig. 8, the wind power in RTED can be represented much more accurately and the number of mixture components of VMD model can be decreased. For the reason that the CDF of actual wind power is obtained by the CDF of forecast error moving to right on the abscissa by $w_{j,t}^{for}$. Wind power EOI fitting method is employed by the CDF fitting for the forecast error CDH between $[m - \bar{w}_{h,j}^{for}, n - \underline{w}_{h,j}^{for}]$, which is in the process of offline look-up table. For instance, the VMD model with three mixture components in Fig. 8 can be simplified to a new VMD model with 1 mixture component in Fig. 9.

For the RTED model with multiple WFs in this paper, each WF can be modeled by VMD by EOI fitting method, resulting in more accurate wind power representation and less mixture components of VMD model. Then the scheduled error of wind power can be reduced and the computational efficiency can be improved. While for the sum WF, only the inverse CDF values $F_{\Sigma,t}^{-1}(1 - c_u)$ and $F_{\Sigma,t}^{-1}(c_d)$ corresponding to the chance constraints in (21) are considered and can be obtained by offline computing of the VMD model of WF-SUM. The number of mixture components of VMD model for WF-SUM has no effect for the RTED model and can be large enough for accuracy. Detailed comparison can be seen in the case study in this paper.

V. RTED Algorithm With Multiple WFs Based on VMD

As analyzed in Chapter IV of this paper, the introduction of VMD makes it possible to employ classical algorithms such as EIFC method and SLP method in dynamic RTED with multiples WFs. The above RTED model with multiples WFs in Chapter IV is solved by classical algorithms in this chapter.

A. The Preparation of SLP Algorithm

The proposed dynamic RTED with multiples WFs can be solved by the widely used SLP algorithm in the power system. However, an important procedure is necessary to set the initial operating point. If the initial operating point deviates from the final optimal solution, the number of iteration algorithms would increase, reducing the efficiency of the algorithm.

If we consider the forecast wind power as the scheduled wind power, the proposed model can be converted to a QP problem [42] and can be fast solved. However, since the PDF curves of wind power forecast error are not symmetrical and the underestimation cost and the overestimation cost coefficient are not same [7], [42], there is a deviation between the scheduled power and forecast power. If the forecast power of WFs is regarded as the scheduled power, the economy will be greatly reduced. This will be proven in the case study of this paper.

While according to this method, if we consider the above forecast wind power and scheduled power of TGs obtained by QP method as the initial operating point for SLP algorithm, the model can be solved. An important defect of this method is that if the forecast wind power deviates from the scheduled wind power, the number of iteration algorithms will increase. To this end, a method to get initial operating point of WFs and TGs named as EIFC-QP algorithm is proposed as follows.

B. Initial Operating Point Solving Method of SLP Algorithm Based on EIFC-QP

For RTED with $T\,(T = 12)$ time interval, the feasible operation regions (FORs) of TGs [28] can be obtained, i.e., the region of generator output reachable from a specified operating point under the constraints (12)–(15), defined as $[\underline{\alpha}_{i,t}, \bar{\alpha}_{i,t}]$. Then in the FORs of TGs and capacities of WFs, respectively, we can get the scheduled power of TGs and WFs in every time interval by EIFC method. As shown in Table I, by the incremental fuel cost of TGs in (19), WFs in (20) and the supply-demand balance in (12), EIFC method can be employed to get the scheduled power of TGs and WFs in every time interval. The scheduled power of WFs is denoted as $w_{j,t}^{0}$ and can be regarded as the initial operating points of WFs in SLP algorithm.

Note that the dispatch method of TGs and WFs is a static ED because the ramp-rate constraint of TGs is not considered

TABLE I
EIFC METHOD OF WFs AND TGs

Generator type	WFs	TGs
Incremental fuel cost	$d_j - k_{un,j} + (k_{un,j} + k_{ov,j})(1 + e^{-\alpha_t (w_{j,t} - \gamma_t)})^{-\beta_t}$	$2a_i p_{i,t} + b_i$
Upper and lower limit	$[0, w_{r,j}]$	$[\underline{\alpha}_{i,t}, \bar{\alpha}_{i,t}]$
Supply-demand balance	$\sum_{i=1}^{I} p_{i,t} + \sum_{j=1}^{J} w_{j,t} = L_t, \forall t$	

in the EIFC method. So the scheduled power of TGs cannot be regarded as the initial operating points of TGs in SLP algorithm. Taking $w_{j,t}^0$ into the above dynamic RTED model, the cost of WFs in (11) changes to be a constant value and the cost of TGs in (10) changes to be a quadratic form. The constants in (12)–(18) remain linear. Then the above dynamic RTED model can be solved by QP method and we can get the scheduled power of TGs in every time interval that is denoted as $p_{i,t}^0$. $w_{j,t}^0$ and $p_{i,t}^0$ can be regarded as the initial operating points of TGs in SLP algorithm, denoted as $P_{temp} = \{p_{i,t}^0, w_{j,t}^0\}$.

C. SLP Algorithm

The procedures of SLP algorithm are as follows [7].

Step 1: In the vicinity of initial operating point P_{temp}, Taylor expansion can be employed for the nonlinear objective function in (24) as shown at the bottom of this page:

The nonlinear objective function in (9) changes to be a linear optimization problem in the user-defined radius of the neighborhood of $P_{GW}^k = \{p_{i,t}^k, w_{j,t}^k\}$. To guarantee the feasible solution of SLP algorithm can be obtained, a load shedding variable ΔL_t is introduced in the linear programming problem (14) and a penalty term $H \cdot \Delta L_t$ is added to eliminate the non-zero load shedding variable finally [7]. We can get the operating points of TGs and WFs that are denoted as $P_{GW}^1 = \{p_{i,t}^1, w_{j,t}^1\}$. The operating points can be regarded as the initial operating points of TGs in the next operating points for the above Taylor expansion, then we can get $P_{GW}^2 = \{p_{i,t}^2, w_{j,t}^2\}$. Repeat this process and we can get $P_{GW}^n = \{p_{i,t}^n, w_{j,t}^n\}$ in the *n*-th times of iteration.

Step 2: Compared with the set accuracy parameters ε and ε', if the scheduled power $\|P_{GW}^n - P_{GW}^{n-1}\|_2 < \varepsilon$, this means that the iterative process tends to be convergent and the iteration stops. If $\|P_{GW}^n - P_{GW}^{n-1}\|_2$ value tends to be stabilized and $\|P_{GW}^n - P_{GW}^{n-1}\|_2 < \varepsilon'$, this means that the iterative process oscillates between two operating points and the iteration stops. Otherwise the iterative process continues and returns to *step 2*, instead P_{GW}^{n-2} by P_{GW}^n until reaching the upper limit of the number of iterations, the iteration stops.

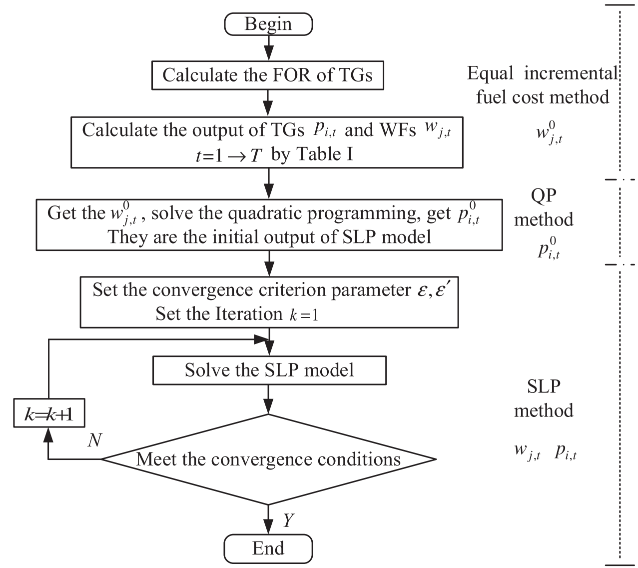

Fig. 10. Framework of the dynamic RTED algorithm with multiple WFs.

D. The Framework of the Proposed Dynamic RTED Algorithm With Multiple WFs

Until now, an EIFC-QP-SLP algorithm is proposed to solve the proposed dynamic RTED model with multiple WFs. Its framework is shown as Fig. 10.

Based on VMD model, RTED model and algorithm with multiple WFs considering the correlation of multiple wind power random variables are proposed. This chapter focuses on how to reduce the number of mixture components (*L*) and increase the representation accuracy of VMD, so as to increase the computation efficiency and scheduled accuracy in RTED.

VI. CASE STUDY

The whole regions of Ireland‘s power system with 53,660 historical data pairs (2014.9.5-2016.3.16) [51] are studied in this chapter. IEEE-30 bus system [56] is employed to test the proposed dynamic RTED model and algorithm with multiple WFs based on VMD. The number of TGs and WFs is 6 and 2, respectively. WF-I and WF-II are connected on the 7th and 21th bus of the system with a capacity of 40 MW and 80 MW, respectively. The total load in the system is 400 MW. *T* is 12 and the time interval is 5 minutes. The RTED creates generation schedules for TGs and WFs within 1 hour (5 min, 10 min . . . 60 min). The parameters of TGs can be found in detail in [56] and the ramp-rate is 1%/min of the capacity. The wind farm operation cost coefficient, underestimation cost and overestimation

$$\min \sum_{t=1}^{T} \begin{bmatrix} H\Delta L_t + \sum_{i=1}^{I} \frac{\partial \sum_{i=1}^{I} C_{g,i,t}(p_{i,t})}{\partial p_{i,t}} \Big|_{p_{i,t}=p_{i,t(k)}} \cdot p_{i,t} \\ + \sum_{j=1}^{J} \frac{\partial C_{w,j,t}(w_{j,t})}{\partial w_{j,t}} \Big|_{w_{j,t}=w_{j,t(k)}} \cdot w_{j,t} \end{bmatrix}. \quad (24)$$

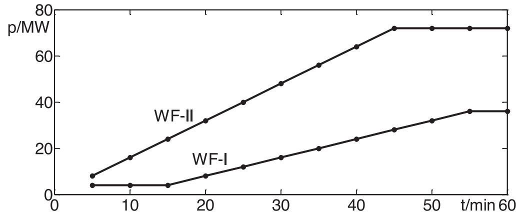

Fig. 11. Forecast power curves of WF-I and WF-II within 1 h.

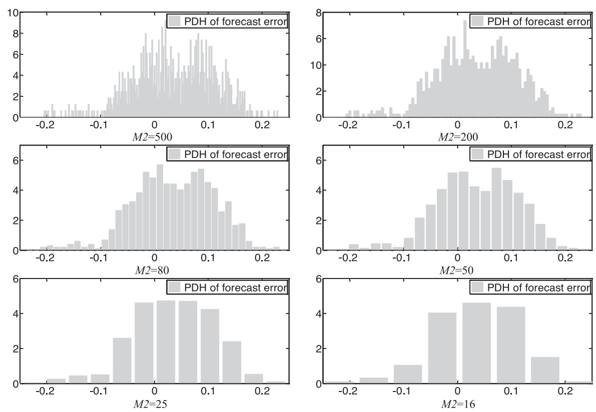

Fig. 12. PDHs of forecast error in the FB 0.55–0.57 p.u. of different *M2*.

cost coefficient is 0\$/MWh, 60\$/MWh and 120\$/MWh, respectively [42]. The confidence levels in RTED, i.e., c_u and c_d, are both 0.95 [54]. The VMD parameters of wind power forecast error are obtained by the whole regions of Ireland's power system that contains forecast wind power and forecast error collected from September 5, 2014 to March 16, 2016 with around 53,660 data pairs [51] and the FBs are designed by proposed IVD bin designing method. WF-I, WF-II and WF-SUM corresponds to Ireland WF, Northern Ireland WF and whole regions of Ireland WF, respectively. The forecast power curves of WF-I and WF-II within 1 hour are shown in Fig. 11. The algorithm is run in MATLAB R2013a on a Core-i5 2.70-GHz notebook computer.

A. Comparison of Representation Effect of PDH by VMD and GMD

The PDHs of forecast error in the FB [0.55-0.57) p.u. of different *M2* are shown in Fig. 12. When *M2* changes from large to small, the feature of two peaks tends to vanish and can be fitted by unimodal distributions such as Gaussian distribution, Beta distribution and IVD. Therefore, large error of PDH representation effect can be obtained with small *M2* and large error of representation effect for wind power will be obtained. So actual power or forecast error PDH accuracy will be reduced if not enough *M2* is used, although the PDH seems to be fitted accurately. Take the forecast error PDH for instance, $M2 = 80$ is chosen for its obvious distribution feature and easy fitting.

The PDH of forecast error in the FB [0.55-0.57) p.u. is fitted by VMD and GMD of different L, $M2 = 80$. Fig. 13 shows the PDF curve of VMD and GMD of $L = 4$ and the PDF curves of their mixture components (IVDs and GDs). Table II and III list the parameters of the above VMD and GMD. Table IV lists the root mean square errors (RMSEs) [7] of the above VMD and GMD (the unit of RMSE is ignored). As shown in Table IV, the RMSEs of VMD of $L = 2, 3, 4$ are 0.2426, 0.2142 and 0.1989, respectively and the RMSEs of GMD of $L = 2, 3, 4$ are 0.2581,

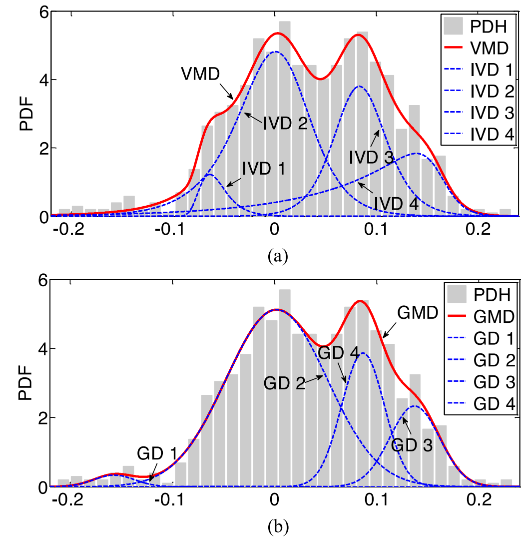

Fig. 13. Comparison of representation effect of VMD and GMD of $L = 4$ for PDH fitting. (a) VMD and its mixture components. (b) GMD and its mixture components.

TABLE II
PARAMETERS OF VMD OF $L = 4$ FOR PDH FITTING

	i of L of VMD			
	$i = 1$	$i = 2$	$i = 3$	$i = 4$
k_i	0.042	0.463	0.261	0.233
α_i	79.44	47.54	55.78	90.05
β_i	39.91	0.678	1.150	0.132
γ_i	−0.110	0.0092	0.081	0.162

TABLE III
PARAMETERS OF GMD OF $L = 4$ FOR PDH FITTING

	i of L of GMD			
	$i = 1$	$i = 2$	$i = 3$	$i = 4$
k_i	0.017	0.637	0.143	0.203
μ_i	−0.157	0.0021	0.137	0.087
σ_i	0.020	0.050	0.024	0.021

TABLE IV
RMSE OF VMD AND GMD FOR PDH FITTING

	RMSE	
L	VMD	GMD
2	0.2426	0.2581
3	0.2142	0.2396
4	0.1989	0.2148

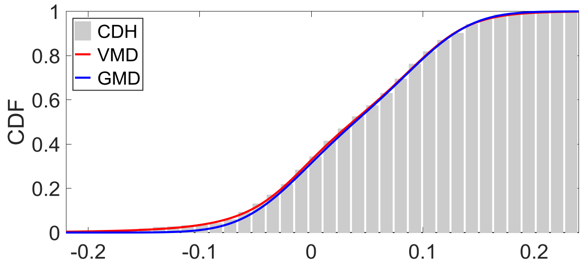

Fig. 14. Comparison of representation effect of VMD and GMD of $L = 2$ for CDH fitting.

TABLE V
PARAMETERS OF VMD OF $L = 4$ FOR CDH FITTING

	i of *L*			
	$i = 1$	$i = 2$	$i = 3$	$i = 4$
k_i	0.017	0.678	0.191	0.114
α_i	139	25.49	55.31	107.9
β_i	0.292	1.789	2.36	0.232
γ_i	−0.153	−0.023	0.071	0.158

TABLE VI
RMSE OF VMD AND GMD FOR CDH FITTING

	RMSE	
L	VMD	GMD
2	0.00282	0.00852
3	0.00236	0.00389
4	0.00179	0.00285

0.2396 and 0.2148, respectively. We can see that compared with GMD, VMD shows better representation effect with the same *L*. And to represent wind power within a certain error tolerance, VMD requires fewer *L* compared with GMD, making up for the lack that IVD has one more parameter than GD.

B. Comparison of Representation Effect of CDH by VMD and GMD

The CDH of forecast error in the FB [0.55-0.57) p.u. is fitted by VMD and GMD of different *L*, $M2 = 80$. Fig. 14 shows the CDF curve of VMD and GMD of $L = 2$. Table V lists the parameters of VMD. Table VI lists the RMSEs of the above VMD and GMD. As shown in Table V, the RMSEs of VMD of $L = 2$, 3, 4 are 0.00282, 0.00236 and 0.00179, respectively and the RMSEs of GMD of $L = 2$, 3, 4 are 0.00852, 0.00389 and 0.00285, respectively. We can see that compared with GMD, VMD shows much better representation effect with the same *L* for CDH fitting.

C. Comparison of Computation Efficiency of VMD and GMD

The average computation times of CDF of VMD and GMD of different *L* in MATLAB are shown in Table VII. The average computation times of the CDF function of VMD of $L = 2$, 3 and 4 are less than 1 microsecond. While the average computation times of the CDF function of GMD are tens of microseconds, almost 80 times that of VMD. Since the CDF of VMD has analytical forms as shown in (5), the computation efficiency is much better than the CDF of GMD with no analytical forms. In addition, to represent wind power within a certain error tolerance, VMD requires fewer *L* compared with GMD. Therefore, VMD has much better computation efficiency for CDF than GMD within a certain error tolerance. It is worth mentioning that tens of microseconds seems not too much, while there may be tens of thousands or even millions of CDF iteration computations in RTED with wind power integration and the computation efficiency is particularly important. The ED model with wind power integration based on VMD can greatly simplify and speed up the ED model and algorithm and this would be discussed in Section VI-E.

TABLE VII
AVERAGE COMPUTATION TIMES OF CDF OF VMD AND GMD OF DIFFERENT *L*

	Time/μs	
L	VMD	GMD
2	0.45	35.8
3	0.61	50.2
4	0.83	68.6

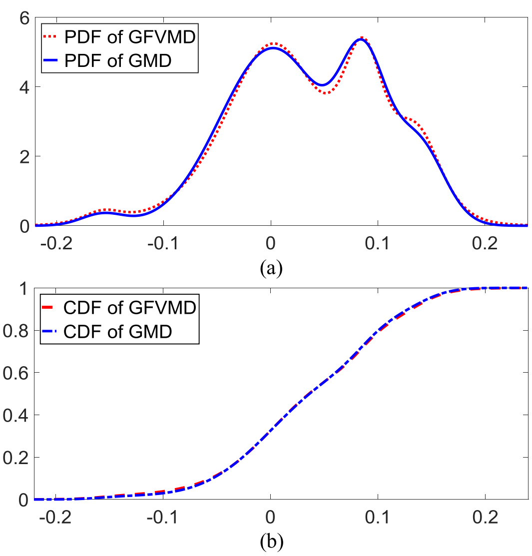

Fig. 15. PDF and CDF curves of GMD of $L = 4$ and the corresponding GFVMD. (a) PDF curve. (b) CDF curve.

D. The Comparison of Representation Effect of CDH by GFVMD and GMD based on GFVMD

The PDF and CDF curves of GFVMD and GMD are as shown in Fig. 15 (a) and (b), respectively. Note that GMD is the same as that in Table III and the corresponding GFVMD (i.e., GFVMD by GMD) is got by parameter relation of GMD (i.e., substituting the parameters α_i, β_i, γ_i and k_i with $1.63/\sigma_i$, 1, μ_i and k_i. The difference of CDF (probability) of the corresponding GFVMD

TABLE VIII
PARAMETERS OF GFVMD OF $L = 4$ FOR CDH FITTING

	i of *L*			
	$i = 1$	$i = 2$	$i = 3$	$i = 4$
k_i	0.0126	0.602	0.279	0.1064
α_i	82.48	33.62	59.89	65.97
γ_i	−0.178	−0.0028	0.0864	0.141

TABLE IX
PARAMETERS OF THE CORRESPONDING GMD OF $L = 4$ BY GFVMD

	i of *L* of GMD			
	$i = 1$	$i = 2$	$i = 3$	$i = 4$
k_i	0.0126	0.602	0.279	0.1064
μ_i	−0.178	−0.0028	0.0864	0.141
σ_i	0.0198	0.0485	0.0272	0.0247

TABLE X
RMSE OF GFVMD FOR CDH FITTING AND THE CORRESPONDING GMD

	RMSE	
L	GFVMD	GMD by GFVMD
2	0.00350	0.00580
3	0.00252	0.00399
4	0.00213	0.00372

compared with GMD is -0.92% $\sim$ 1.02%. And in the middle part of CDF, the two curves almost completely overlap. So in practical engineering, the corresponding GDVMD can approximate or substitute GMD in high accuracy.

We can also fit the CDH in Fig. 14 by GFVMD and the parameters of GFVMD are as shown in Table VIII. By parameter relation (i.e., substituting the parameters α_i, β_i, γ_i and k_i with $1.63/\sigma_i$, 1, μ_i and k_i) we can get the corresponding GMD and its parameters are shown in Table IX. The RMSEs of GFVMD and the corresponding GMD (i.e., GMD by GFVMD) of different *L* are shown in Table X.

Compared with VMD in Table IV and GFVMD in Table X of same *L*, we can see that GFVMD shows worse representation effect. The reason is that the parameter β is constrained to 1 in GFVMD and the ability of distortion is lost. The corresponding GMD (i.e., GMD by GFVMD) shows worse representation effect compared with GFVMD for the difference of CDF (probability) of the corresponding GMD compared with GFVMD.

E. The Result of the Proposed RTED Model and Algorithm Based on VMD and GMD

To test the proposed RTED model and algorithm based on VMD and GMD and the wind power EOI fitting method, the scheduled power of TGs and WFs of the proposed RTED model and algorithm based on the following four wind power representation models are compared: 1) actual CDH of WFs, i.e., the accurate representation of WFs; the errors of the actual CDH and other distribution models are the accuracy of other distribution models; 2) VMD model; the minimum number of mixture components is the number to make the CDF fitting error less than 0.5% and the maximum number of mixture components is 3; 3) GMD model; the number of mixture components of GMD is same with the number of mixture components of VMD in the same forecast bin; 4) VMD model by wind power EOI fitting method; the number of mixture components is 1.

The scheduled power of WF-1 and WF-II based on four wind power representation models is shown in Fig. 16. The total cost of 1 hour based on four wind power representation models is shown in Table XI. Table XI also shows the total cost that the wind power is regarded as "non-scheduled", i.e., take forecast wind power as scheduled wind power. Table XII compares the

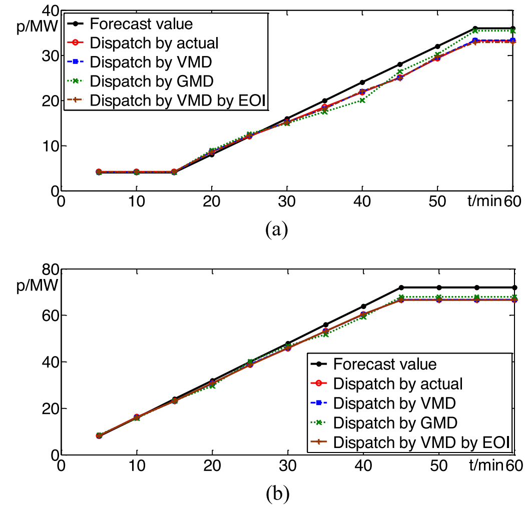

Fig. 16. Forecast power and scheduled power based on actual CDH, VMD, GMD, and VMD by EOI models of WFs. (a) WF-I. (b) WF-II.

TABLE XI
THE TOTAL COST OF 1 HOUR BASED ON ACTUAL CDH, VMD, GMD AND VMD BY EOI MODELS OF WFS AND NON-SCHEDULED STRATEGY

	TGs/$	WFs/$	Total/$
Actual	21046.7	770.2	21816.9
VMD	21041.9	780.1	21822.0
GMD	21068.7	794.4	21863.1
VMD by EOI	21048.5	781.2	21829.7
Non-scheduled	20795.4	1339.7	22135.1

TABLE XII
THE COMPUTATION TIME OF THE PROPOSED RTED ALGORITHM BASED ON THE FOUR WIND POWER REPRESENTATION MODELS

	Calculation time/s
Actual	7.456
VMD	1.741
GMD	4.125
VMD by EOI	1.583

computation time of the proposed RTED algorithm based on the four wind power representation models.

As shown in Fig. 16, the scheduled power of WF-1 and WF-II based on actual CDH of WFs (the red solid lines) can be regarded as the accurate scheduled result theoretically. Since the PDF curves of wind power forecast error of WFs are not symmetrical and the underestimation cost and the overestimation cost coefficient are not same, there is a deviation between the scheduled power and forecast power of WF-1 and WF-II. If the forecast powers of WFs are regarded as the scheduled power (non-scheduled), the economy will be greatly reduced as shown in Table XI. Compare the scheduled power of WFs based on VMD and GMD models, since VMD has better representation effect for wind power compared with GMD with same number of mixture components, the scheduled power based on VMD model (the blue dashed lines) is more closed to the scheduled power based on actual CDH than the scheduled power based on GMD model (the green dotted lines), resulting in better economy as shown in Table XI.

Since the scheduled power of WF-I and WF-II based on GMD model has more deviation compared with the scheduled power, there is an additional cost of \$46.2 per hour. If the wind forecast power and load remain the same, the additional cost is \$404,712 per year.

Although the number of mixture components of VMD model by wind power EOI fitting method is only 1, the VMD model by wind power EOI fits the wind power CDH that is related to the economic principle in RTED, resulting in similar scheduled accuracy as shown in Fig. 16 (the brown dash-dot lines) and similar system cost as shown in Table XI compared with VMD model with three mixture components.

Table XII shows the computation time of the proposed RTED algorithm based on the four wind power representation models. We can see that the RTED algorithm based on VMD has much better computation efficiency than the algorithm based on GMD, showing an obvious advantage in RDED that has strict requirements for computation time. The RTED algorithm based on actual CDH shows the longest computation time among the four wind power representation models. Since wind power EOI fitting method reduces the number of mixture components of VMD model, VMD model by wind power EOI fitting method improves the computation efficiency of the RTED algorithm based on VMD.

F. The Result of the Proposed RTED Model and Algorithm Based on GD, BD and IVD

The scheduled power of TGs and WFs of the proposed RTED model and algorithm based on the classical distribution models (GD, BD and IVD) are compared: 1) wind power of WF-I, WF-II and WF-SUM are modeled by GD model; 2) wind power of WF-I, WF-II and WF-SUM are modeled by BD model; 3) wind power of WF-I, WF-II and WF-SUM are modeled by IVD model proposed in Part I of this paper.

As shown in Fig. 17, the scheduled power of WF-I and WF-II based on GD, BD and IVD model is the blue dashed lines, green dotted lines and brown dash-dot lines, respectively.

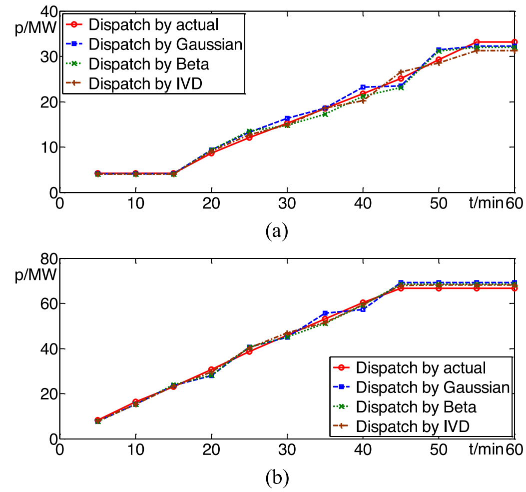

Fig. 17. Scheduled power based on actual, GD, BD and IVD models of WFs. (a) WF-I, (b) WF-II.

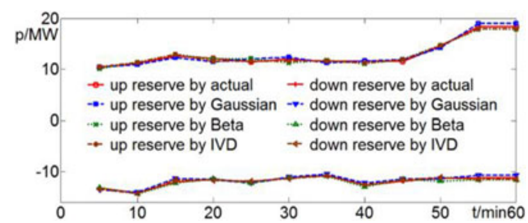

Fig. 18. Required regulation reserves accuracy of RTED model based on GD, BD, and IVD models of WFs.

TABLE XIII
THE TOTAL COST OF 1 HOUR BASED ON GD, BD AND IVD MODELS OF WFS

	TGs/\$	WFs/\$	Total/\$
GD	21062.6	865.3	21927.9
BD	21062.7	857.3	21920.0
IVD	21058.6	842.5	21901.1

For the convenience of comparison, the accurate scheduled wind power based on actual CDH of WFs (the red solid lines) is also shown in Fig. 17. Fig. 18 compares the required regulation reserves of the system for wind power based on the above three distribution models. For the convenience of comparison, the up required regulation reserves and down required regulation reserves are distinguished by a positive and negative value. The total costs of 1 hour based on above three models are shown in Table XIII.

As shown in Fig. 17, there are large deviations between the accurate scheduled power of WF-I and WF-II (based on actual CDH) and scheduled power based on GD, BD and IVD. This is because wind power cannot be represented accurately by the above distribution models, reducing the economy of the power system. In addition, the wind power that corresponds to c_u (i.e., $F_{\Sigma,t}^{-1}(1-c_u)$) and c_d (i.e., $F_{\Sigma,t}^{-1}(c_d)$) is not represented accurately by GD, BD and IVD, reducing the required regulation reserves accuracy. As shown in Fig. 18, take down required regulation reserves for instance, if the value of $F_{\Sigma,t}^{-1}(c_d)$ obtained by the above three distribution models is greater than the actual value, it means that unnecessary additional reserves are left, resulting in worse economy. If the value of $F_{\Sigma,t}^{-1}(c_d)$ obtained by the above three distribution models is less than the actual value, it means that no enough reserves are left, bringing more risk to the power system.

Therefore, classical unimodal distribution models such as GD, BD and IVD models cannot represent wind power accurately, greatly reducing the economy of the power system. As shown in Table XIII, there is an additional cost of 111$, 103.1$ and 84.2$ per hour if wind power is represented by GD, BD and IVD model, respectively. If the wind forecast power and load remain the same, the additional cost is 972 360$, 903 156$ and 737 592$ per year.

G. The Result of the Proposed RTED Based on Different Regulation Reserves Strategies

As analyzed in Section II-B, the proposed dynamic stochastic RTED model and algorithm with multiple WFs based on VMD can consider the effect of correlation of multiple WFs to the system regulation reserves compared with other regulation reserves strategies. WF-I and WF-II are modeled by VMD model and the following three regulation reserves strategies are compared.

1) System regulation reserves are set by the proposed strategy, i.e., the distribution of WF-SUM represented by VMD model in (13), denoted as "WF-SUM strategy".
2) System regulation reserves are set by certain proportion of the WF capacity [54]–[55], denoted as "determined strategy". To set enough regulation reserves, the up regulation reserve capacity and down regulation reserve capacity are consistent with the largest up regulation reserve capacity and down regulation reserve capacity in whole time interval by actual CDH in Fig. 18. That is to say 15.27% up regulation reserve capacity and 11.84% down regulation reserve capacity of the whole WF capacities (120 MW) are set, respectively.
3) System regulation reserves are set by the wind power distributions of each WF respectively, denoted as "respective strategies." To set enough system regulation reserves in chance constrains in (9), the confidence levels of each WF are both 0.95. Take down required regulation reserves for instance, that means that value of $F_{\Sigma,t}^{-1}(c_d)$ can be obtained by weighted average calculation of $F_{\mathrm{WF-I},t}^{-1}(c_d)$ and $F_{\mathrm{WF-II},t}^{-1}(c_d)$.

The required regulation reserves based on the above three regulation reserves strategies are shown in Fig. 19. The total cost of 1 hour based on above three regulation reserves strategies are shown in Table XIV.

As shown in Fig. 19, since the determined strategies (the blue dashed lines) widely used in robust ED have to set enough reserve for every time interval by same regulation reserve capacities, unnecessary additional reserves are left. Take the up regulation reserve, more regulation reserve capacities are required in 55 and 60 minute time interval and the required regulation reserve capacities are set for the overestimation of wind power. While in other time intervals of RTED, unnecessary additional reserves are left since the required regulation reserve capacities are less. Then TGs are deviated from the point of economic operation, reducing the economy of the power system. For respective strategies, since the correlation of multiple WFs is not considered, unnecessary additional reserves are left as shown in Fig. 19 (green dotted lines). Then TGs are also deviated from the point of economic operation, reducing the economy of the power system. As shown in Table XIV, if the determined strategies and respective strategies are employed, there are additional cost of 60.1$ and 23.4$ per hour, respectively compared with the proposed WF-SUM strategy. If the wind forecast power and load remain the same, the additional cost is 526 476$ and 204 984$ per year.

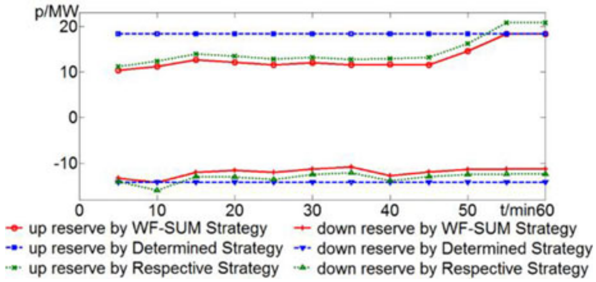

Fig. 19. Required regulation reserves accuracy of RTED model based on three regulation reserves strategies.

TABLE XIV
THE TOTAL COST OF 1 HOUR BASED ON THREE REGULATION RESERVES STRATEGIES

	TGs/$	WFs/$	Total/$
WF-SUM Strategy	21041.9	780.1	21822.0
Determined Strategy	21088.8	793.3	21882.1
Respective Strategy	21055.7	789.7	21845.4

VII. CONCLUSION

IVD increases the representation accuracy of wind power and has the same and mathematical analytical form compared with VD. Based on IVD, the proposed VMD can represent all possible wind power distribution accurately. The case study shows the advantages of better representation accuracy and computation efficiency compared with other classical distribution models.

The proposed stochastic dynamic RTED model with multiple WFs based on VMD can describe the correlation between multiple WFs on the system regulation reserves accurately. The case study shows that compared with GMD model and classical unimodal distribution models such as GD, BD and IVD, the integration of VMD model has better computation efficiency and can reduce the error of scheduled power and system regulation reserves, improving the system economy. The proposed regulation reserves strategies based on the representation of the sum WF can consider the correlation between multiple wind power random variables. Compared with other classical reserves strategies, the proposed strategies can set the required system regulation reserves more accurately.

The confidence levels relating to the chance constraints are regarded as an constant value in this paper. The potential economy would be increased by the optimization of confidence levels. The following studies would regard confidence levels as optimized value.

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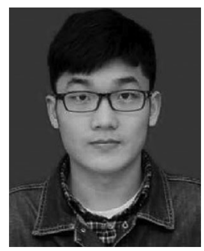

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